[Project Title]

IEOR 242 Spring 2026 Final Project Report

Team Information Withheld in Public Repository

Spring 2026

This public repository version is a clean report template. Prior course identifiers and participant names have been removed. Replace the bracketed prompts below in your private submission copy.

Abstract

Summarize the problem, dataset, main deep learning method, one advanced element beyond class material, strongest result, and one sentence on limitations. Keep this to roughly 150–250 words.

[Write abstract here.]

1 Project Description and Motivation

Clearly identify the application, the use-case, and why the problem matters in its domain. Explain why a deep learning approach is appropriate instead of a simpler baseline alone.

Problem statement. [State the learning task in one concise paragraph.]

Motivation. [Explain the domain relevance, practical value, and why the problem is conceptually interesting.]

Project goals.

Goal 1 [State the main technical objective.]

Goal 2 [State the evaluation or deployment-oriented objective.]

Goal 3 [State the scientific question, ablation, or comparison you want to answer.]

2 Data and Learning Problem

Describe the dataset clearly enough that a reader can understand the inputs, outputs, and prediction target without seeing the code. Discuss exploration, labeling, preprocessing, and any quality issues.

2.1 Dataset Overview

- Dataset source: [Name, citation, or URL]
- Domain: [Finance / vision / language / multimodal / other]
- Number of samples: [Insert count]
- Input representation: [Features, dimensionality, sequence length, image size, etc.]
- Output / labels: [Classes, regression target, structured output, etc.]

2.2 Learning Task Definition

Make the supervised, unsupervised, or self-supervised learning problem explicit. If supervised, define X, Y, and the mapping you are trying to learn.

Let X denote [define inputs] and let Y denote [define labels or targets]. Our task is to learn a mapping

$$f_\theta : X \mapsto Y$$

that optimizes performance on [state metric or task objective].

2.3 Data Preparation and Quality Considerations

- Preprocessing pipeline: [normalization, tokenization, feature engineering, augmentation, filtering, etc.]
- Train / validation / test split: [chronological, random, grouped, cross-validation, etc.]
- Data quality issues: [missing values, duplicates, noisy labels, outliers]
- Class balance: [balanced / imbalanced, with counts or proportions]
- Leakage prevention: [steps taken to ensure causal or proper separation]

3 Methodology

This section should justify the architecture, objective, optimizer, and regularization strategy. You must also describe at least one advanced methodological element beyond what was covered in class.

3.1 Model Architecture

Describe the network in enough detail that another student could reimplement it: layers, hidden dimensions, nonlinearities, normalization, attention, recurrence, or other structural choices.

[Describe your baseline model and your final model here.]

3.2 Objective Function and Training

We train parameters θ by solving

$$\min_\theta \frac{1}{N} \sum_{i=1}^{N} L(f_\theta(X_i), Y_i) + R(\theta).$$

- Loss L: [cross-entropy, MSE, focal loss, contrastive loss, etc.]
- Optimizer: [mini-batch SGD, Adam, AdamW, etc.]
- Learning-rate schedule: [constant, cosine decay, warmup, step decay, etc.]
- Regularization $R(\theta)$: [L_2, dropout, data augmentation, early stopping, label smoothing, etc.]
- Batch size / epochs: [insert values]

3.3 Advanced Component Beyond Class Material

This is required. Examples include a newer architecture block, a specialized loss, transfer learning protocol, calibration method, uncertainty estimation, diffusion-style component, LoRA-style fine-tuning, a domain-specific augmentation, or another method not covered directly in lecture. Explain both what it is and why it helps for your problem.

[Describe the advanced component here, including its implementation details and rationale.]

3.4 Baselines and Comparisons

Explain what you compare against and why those comparisons are fair. At minimum, include one simpler baseline and one comparison that tests whether the advanced component helps.

- Baseline 1: [simple model or heuristic]
- Baseline 2: [course-standard deep model]
- Proposed model: [your final method]
- Ablation: [remove advanced component / change loss / freeze layers / etc.]

4 Experimental Setup

Provide enough detail for reproducibility: hardware, software stack, hyperparameter selection strategy, and the exact evaluation metrics.

- Hardware / runtime: [GPU type, CPU, training time]
- Software: [PyTorch / TensorFlow / JAX version, key libraries]
- Hyperparameter selection: [grid search, manual tuning, validation set]
- Primary metrics: [accuracy, F1, AUROC, RMSE, BLEU, etc.]
- Secondary diagnostics: [confusion matrix, calibration error, loss curves, qualitative outputs]

5 Results and Discussion

Present both quantitative results and diagnostic analysis. You are expected to show training and validation loss curves and interpret them in terms of underfitting or overfitting.

5.1 Main Results

Table 1: Placeholder table for the main experimental results. Replace with your actual metrics.

Model	Validation Metric	Test Metric	Notes
Baseline 1	[ ]	[ ]	[ ]
Baseline 2	[ ]	[ ]	[ ]
Proposed model	[ ]	[ ]	[ ]
Ablation	[ ]	[ ]	[ ]

5.2 Training and Validation Curves

Figure 1: Insert training and validation loss curves here. Discuss whether the model appears underfit, overfit, or well-regularized, and relate that diagnosis to your methodological choices.

5.3 Interpretation

- What worked best, and why?
- Where did the model fail or remain unstable?
- Did the advanced component improve results? If so, under what conditions?
- Is the final performance practically useful for the motivating use-case?

A scientifically valid conclusion can still be negative. If performance is insufficient, explain why the result is still informative.

[Write the main analysis here.]

6 Safety, Security, and Ethics

Assume the model is deployed at scale. Discuss potential harms, failure modes, misuse risks, privacy concerns, fairness issues, security vulnerabilities, and any domain-specific regulatory concerns. Tie the discussion back to your actual task rather than writing generic AI ethics text.

- Safety risks: [false positives / false negatives / unsafe recommendations / automation bias]
- Security risks: [data poisoning, model inversion, prompt injection, adversarial manipulation, leakage]
- Ethics concerns: [fairness, representational bias, consent, surveillance, unequal impact]
- Mitigations: [human review, calibration thresholds, monitoring, audits, narrower deployment scope]

[Write the synthesized team discussion here.]

7 Presentation Summary

The course requires a short recorded presentation. Summarize the structure of the talk and optionally include a private link in the submission copy.

1. Problem formulation and motivation.
2. Dataset and preprocessing.
3. Model architecture and advanced component.
4. Main results and loss-curve diagnosis.
5. Risks, limitations, and next steps.

8 Conclusion

Conclude with the main technical takeaway, the strongest result, the main limitation, and one concrete future direction.

[Write conclusion here.]

Submission Checklist

- Clear project description and motivation.
- Dataset exploration and precise learning problem statement.
- Architecture, activation functions, objective, optimizer, and regularization explained.
- At least one advanced methodological component beyond course coverage.
- Training and validation loss curves included and interpreted.
- Final metrics discussed with an honest assessment of practical usefulness.
- Safety, security, and ethics discussion included.
- Short presentation prepared separately.